Part – D Answer the following questions in about a page:

1. A Research scholar researching the market for fresh cow milk assumes that $Qt = f(Pt, Y, A, N, Pc)$ where Qt is the quantity of milk demanded, Pt is the price of fresh cow milk, Y is average household income, A is advertising expenditure on processed pocket milk, N is population and Pc is the price of processed pocket milk .
   - **(a)** What does $Qt = f(Pt, Y, A, N, Pc)$ mean in words?
   - **(b)** Identify the independent variables.
   - **(c)** Make up a specific form for this function. (Use your knowledge of Economics to deduce whether the coefficients of the different independent variables should be positive or negative.)
2. Calculate the elasticity of demand for the demand schedule by using differential calculus method $P = 60 - 0.2Q$ where price is (i) zero, (ii) ₹20, (iii) ₹40.
3. The demand and supply functions are $p_d = 1600 - x^2$ and $p_s = 2x^2 + 400$ respectively. Find the consumer's surplus and producer's surplus at equilibrium point.
4. What are the ideas of information and communication technology used in economics?

ACTIVITY

1. The petrol consumption of your car is 16 Kilometers per litre. Let x be the distance you travel in Kilometers and p the price per litre of petrol in Rupees. Write expressions for demand for Petrol.
2. Make up your own demand function and then derive the corresponding MRfunction and find the output level which corresponds to zero marginal revenue.
3. Use an Excel spreadsheet to calculate values for Quantity of demand at various prices for the function Q = 100−10P then plot these values on a graph.
4. Open MS-Word and put the title as **PRESENT AND ABSENT OF STUDENTS** and insert the table and collect the data for all classes of your school and find the class of highest absentees in a month. Justify with reason for the absentees in a paragraph by using MS Word.

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